

Divide & Conquer Homework

In this assignment you will use the provided code template to implement solutions to each given problem. The solutions you submit must follow the guidelines provided in this document as well as any given in code.

Your solutions must implement a divide & conquer approach to each given problem. Both recursive and iterative implementations are accepted.

When using a known divide & conquer algorithm, only those covered in course material (see Ed) are allowed, and some are provided for you to use.

Restrictions

- The use of built-in `sort` capabilities are strictly prohibited.
- Template code is provided and must be used.
 - ***Note: The templates are refreshed each semester. Be sure to use the current version.***
- Your code must be compatible with **Python 3.10**.
- Do not add imports for any libraries beyond the Python standard library.
- Do not modify the function name or definition.

Testing

For each problem, one or more base test cases are provided. A properly implemented solution will pass any provided base test cases, but their trivial nature is not intended to ensure the overall correctness of your algorithm.

These are the same test cases that will be used to provide assurance that your submitted solutions run using the autograder. We do not release any other test cases.

In problem instances where more than one solution is optimal, *any* optimal solution will be considered correct, assuming that is the solution your algorithm produces.

The test cases can - and should! - be expanded while developing your solutions.

You can run your test cases by issuing the `python3 -m unittest` command from the directory containing your solution files.

Submission

Submit only - and all of - your solution files (i.e., `cs6515_[problem_name].py`) to the Gradescope assignment on or before the posted due date. Do not submit a zip file or any other files except those matching the naming convention defined.

You may submit your solutions as many times as desired and the most recent submission is what will be graded. Execution of your code may take up to 60 minutes. If not completed before the due date, your previous submission will be used.

Faster and correct solutions are worth more credit.

After your submission completes execution, you will see a score of - / 20 listed until grading is completed.

Raquel's Blocks

Brito is playing with his daughter Raquel. He has n blocks with blue numbers and she has m blocks with red numbers.

Together, they are each building a tower to see who can make the tallest one.

While doing so, Brito notices something interesting. Each of the towers are being built such that the numbers are ordered. Brito's in ascending order and Raquel's in descending order, from the ground up.

Brito challenges Raquel to determine the median number out of the blue and red numbers being used to build the towers. If she can do it before Brito she gets a treat!

Develop an efficient divide & conquer algorithm to determine the median of the two block towers.

Notes:

- $m \geq 2$ and $n \geq 2$
- When calculating a median of a data set, it should match the value produced by [statistics.median_low](#)

Implement your algorithm for solving this problem in **cs6515_raquels_blocks.py**.